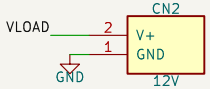
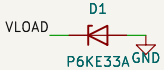
ESP32-C3 SuperMini Multiswitch: 8 Kanäle (ULN2803A) + CRSF – Entwurf v0.3

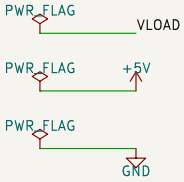
Last: max. 24V / 500mA je Kanal. ESP32 wird aus +5V von CN1 (BEC/Empfaenger) versorgt. Adresse per Firmware/Web-UI/Lua. Keine Pulldowns, keine UART-Vorwiderstaende. R9-R16 und R1-R8: Standard 150R, je Verbraucher waehlen.

UART A = zum Empfaenger bzw. vorigen Modul (Pin3 RX, Pin4 TX). UART B = zum naechsten Modul (Pin3 TX, Pin4 RX): 1:1-Kabel genuegt.

Lastversorgung 12-24V (VLOAD)

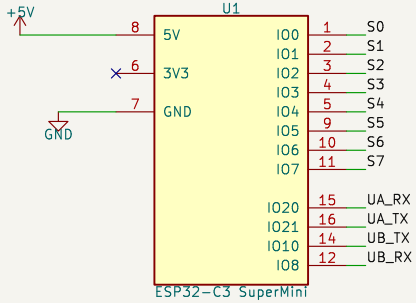




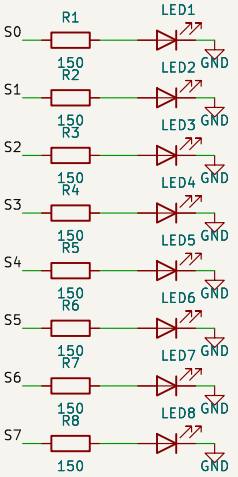


D1: TVS Anode GND, Kathode VLOAD (P6KE33A: ok bis 24V)

8-Kanal-Lastschalter (low-side), Eingaenge direkt vom ESP32

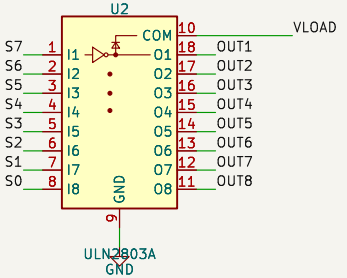


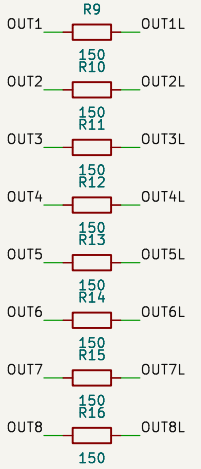
Kontroll-LEDs (Sn -> R -> LED -> GND)

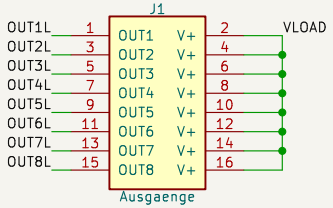


J1: ungerade Pins = OUT1..8 (ueber R9-R16), gerade Pins = VLOAD

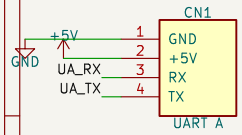
Ausgangswiderstaende (THT, je Verbraucher waehlen)



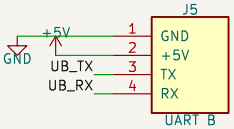




UART A (Empfaenger / vorheriges Modul)



UART B (naechstes Modul)



Entwurf – vor Layout/Bestellung pruefen

Sheet: /
File: CRSF-Multiswitch-ESP32-C3.kicad_sch

Title: CRSF Multiswitch ESP32-C3 SuperMini (ULN2803A)

Size: A4	Date: 2026-10-14	**Rev: v0.3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1